

seed 9 | mass x1.58 friction x0.78 | plate r=5.3cm mug r=1.4cm cutlery 9.7cm | light z=1.40m



Bimanual SO-101: LLM-Planned Table Setting

MuJoCo · OpenVINO INT4 planner + INT8 perception · 10/10 randomized seeds

Problem

Intel challenge: two SO-101 arms in MuJoCo
set a dinner table from a language command,
reason over camera images, coordinate both
arms, and run optimized on Intel hardware.

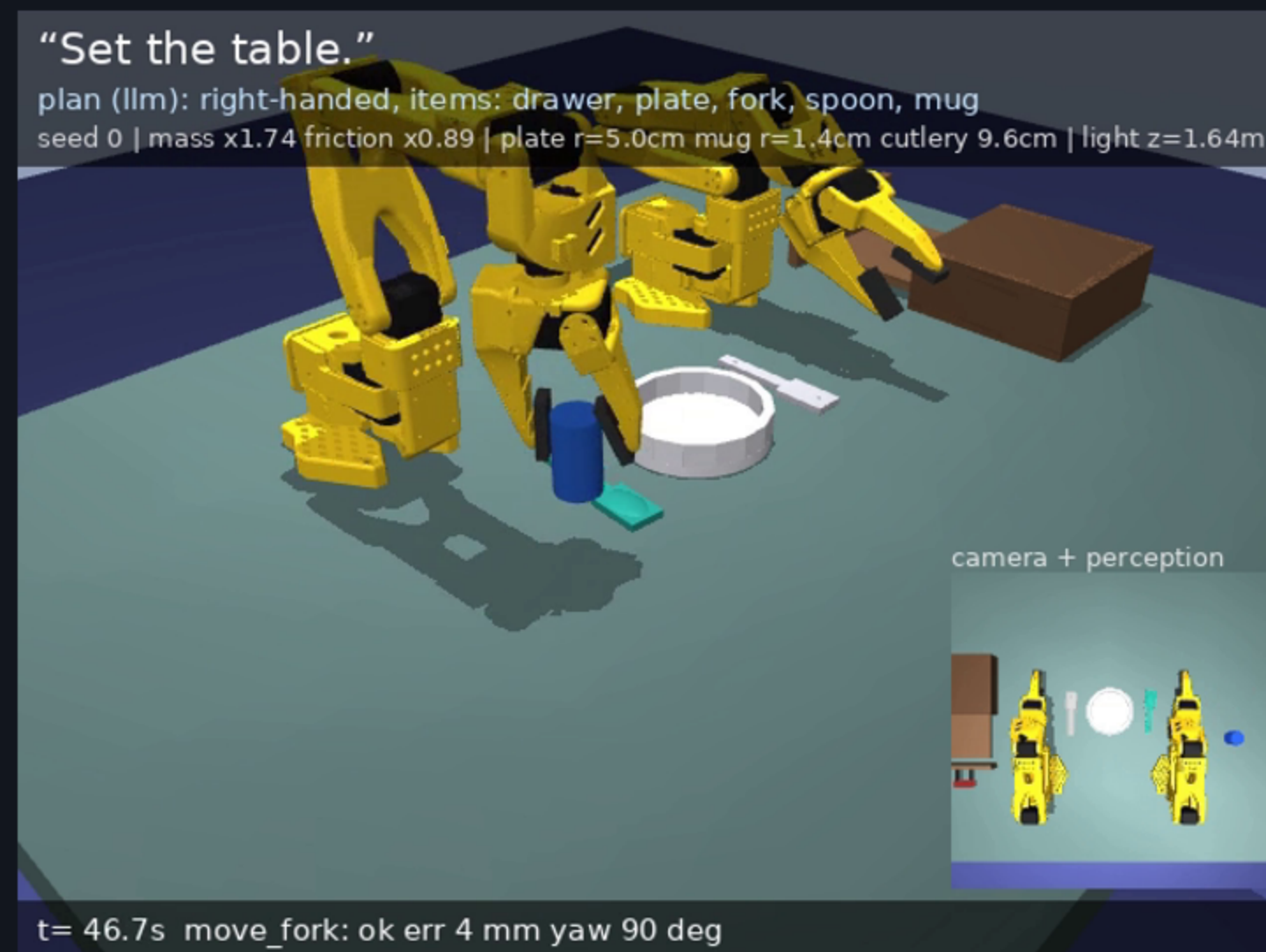
Task: open drawer, retrieve spoon + fork,
hand-off, place plate, cutlery and mug.



Architecture

- 1 Understand: Qwen2.5-1.5B-Instruct INT4
via OpenVINO GenAI -> JSON plan
- 2 Observe: overhead camera -> U-Net seg,
OpenVINO INT8 (~2.3 ms/frame CPU)
- 3 Act: DLS-IK skills for both arms;
hand-off via shared staging zone

Object poses come from the camera.



Results

Tuning seeds: 10 / 10

Hold-out seeds: 10 / 10

Left-handed layout: 10 / 10

Planner valid plans: 12/14 (rules 11/14)

Randomized: mass, friction, size,
lighting, table colour, placement.

Per-step placement error: 1-9 mm.



Intel optimization & honest scope

OpenVINO: planner INT4 (GenAI), perception INT8 IR.

Cloud CPU bench: segnet 2.3 ms, planner 2.8 s/plan, MuJoCo 20.4k steps/s.

Bench script reports latency, throughput, device and precision for CPU / iGPU / NPU.

Learned: planner + perception. Scripted: IK manipulation skills.

Not yet done: Core Ultra run, distilled policy, pouring.